RIFAT ERDEM SAHIN

AI ENGINEERING LEAD | LLM SOLUTIONS ARCHITECT

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📅 **Schedule a Call:** https://calendly.com/rifaterdem/schedule

PROFESSIONAL SUMMARY

AI Engineering Lead specializing in **Generative AI, Large Language Models, and production-grade AI systems**. Deep hands-on experience architecting and deploying enterprise LLM applications with focus on **RAG architectures, AI security, and scalable deployment**. Proven track record building AI-powered automation systems that delivered 300% productivity improvements and 30% cost reductions. Expert in bridging the gap between cutting-edge AI research and production-ready, secure, enterprise solutions.

CORE AI COMPETENCIES

🤖 **Generative AI & LLM Engineering** - Deep expertise building and deploying production Generative AI solutions - Extensive hands-on experience developing Large Language Model applications - Advanced prompt engineering, fine-tuning, and model optimization - Expert in LangChain, LlamaIndex, and modern AI orchestration frameworks

🔄 **RAG Architecture & Vector Systems** - Advanced implementation of Retrieval Augmented Generation (RAG) architectures - Expert in vector databases (Pinecone, Weaviate, ChromaDB, FAISS) - Document processing pipelines and semantic search optimization - Hybrid search strategies combining dense and sparse retrieval

🔒 **AI Security & Governance** - Expert-level knowledge of AI security practices and threat models - Prompt injection prevention and adversarial attack mitigation - Data privacy controls, PII detection, and secure model deployment - Compliance frameworks for regulated industries (finance, healthcare)

🏗️ **AI System Architecture** - Track record shaping enterprise AI architecture decisions - Scalable, maintainable AI solutions with proper observability - MLOps pipelines for continuous model deployment and monitoring - Cost optimization strategies for LLM inference at scale

🧪 **AI Testing & Quality Assurance** - Comprehensive testing approaches: unit, integration, end-to-end - LLM evaluation frameworks and metrics (RAGAS, LangSmith) - A/B testing strategies for prompt and model optimization - Automated quality gates for AI system releases

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KEY AI ACCOMPLISHMENTS

🏆 2026 – Present | Delivery Pilot — Skool Platform

AI Professional Transformation & Forward Deployed Engineering Mentorship - Mentoring engineers and technology professionals to transition into AI Engineers and Forward Deployed Engineers (FDE) by closing practical and technical skill gaps. - Designing and delivering hands-on curriculum on the Skool Platform covering AI-augmented engineering, RAG architectures, multi-LLM orchestration, agentic workflows (n8n, MCP), and GitOps delivery. - Guiding cohort members through building, evaluating, and deploying production-grade AI applications and cloud-native integrations. - **Technologies:** Anthropic Claude, OpenAI, RAG, Qdrant, n8n, LangChain, Kubernetes, Docker, Python, Skool Platform

🏆 2025 | IBM | London, UK

Enterprise AI & Hybrid Cloud Transformation - **Challenge:** Lead enterprise-scale hybrid cloud transformation and AI integration for global clients - **Solution:** Architected comprehensive IBM Cloud + Red Hat OpenShift platform with watsonx AI integration and DevSecOps practices - **Impact:** - 35% reduction in operational overhead through AI-driven automation with watsonx - 40% improvement in deployment frequency via DevSecOps and Ansible Automation Platform - Zero-downtime migration of mission-critical workloads to hybrid cloud architecture - **Technologies:** IBM Cloud, Red Hat OpenShift, watsonx AI, Ansible Automation Platform, Terraform, Kubernetes, GitHub Actions

🏆 2024 | Goldman Sachs | Muscat, Oman

AI-Driven CI/CD Automation Framework - **Challenge:** Manual deployment processes causing bottlenecks and errors in enterprise fintech environment - **Solution:** Architected and deployed AI-powered CI/CD framework using LLMs for intelligent automation - **Impact:** - 300% increase in deployment frequency - 30% reduction in operational costs - 85% reduction in deployment-related incidents - **Technologies:** Python, LangChain, OpenAI GPT-4, Azure OpenAI, Kubernetes, GitLab CI/CD - **AI Components:** Custom RAG system for documentation, automated code review agents, intelligent error detection

🏆 2023 | Ypsomed | Switzerland

IoT Data Intelligence & Predictive Systems - **Challenge:** Complex IoT device data requiring intelligent analysis and predictive maintenance - **Solution:** Implemented ML-powered analytics pipeline with automated anomaly detection - **Impact:** - 40% reduction in versioning conflicts through intelligent merge strategies - Predictive maintenance reducing device failures by 35% - **Technologies:** Python, TensorFlow, Azure ML, IoT Hub, Time-series analysis - **AI Components:** Anomaly detection models, predictive maintenance algorithms, automated data quality checks

🏆 2022 | Cushman & Wakefield | London, UK

Intelligent ETL & Data Pipeline Automation - **Challenge:** Manual data processing creating bottlenecks in real estate analytics - **Solution:** Built AI-enhanced ETL pipelines with intelligent data validation and transformation - **Impact:** - 50% increase in data processing speed - 90% reduction in data quality issues - Real-time analytics enabling faster business decisions - **Technologies:** Python, Apache Airflow, Azure Data Factory, ML-based data validation - **AI Components:** Automated data quality scoring, intelligent schema mapping, anomaly detection in data flows

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TECHNICAL EXPERTISE

AI/ML Frameworks & Tools

LLM Platforms: OpenAI API (GPT-4, GPT-3.5), Anthropic Claude, Azure OpenAI Service, AWS Bedrock
Orchestration: LangChain, LlamaIndex, Semantic Kernel, Haystack
Vector Databases: Pinecone, Weaviate, ChromaDB, FAISS, Qdrant, Milvus
ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers
Fine-tuning: LoRA, QLoRA, PEFT, Supervised Fine-Tuning (SFT)
Evaluation: RAGAS, LangSmith, Weights & Biases, MLflow

AI Development & MLOps

Languages: Python (Advanced), JavaScript/TypeScript, Bash, SQL
AI Libraries: OpenAI Python SDK, LangChain, pandas, numpy, sentence-transformers
MLOps: MLflow, Kubeflow, Azure ML, AWS SageMaker, Model versioning
Experiment Tracking: Weights & Biases, MLflow, TensorBoard
Data Processing: Apache Spark, Pandas, Dask, Apache Airflow

Cloud & Infrastructure

Cloud Platforms: Azure (Azure OpenAI, ML, Functions), AWS (Bedrock, Lambda, SageMaker), GCP
Containers: Docker, Kubernetes, Helm (for ML workloads)
Infrastructure as Code: Terraform, Ansible
CI/CD: Jenkins, GitLab CI, GitHub Actions, Azure DevOps
Monitoring: Prometheus, Grafana, ELK Stack, Application Insights

AI Security & Compliance

Security Clearances: UK SC (valid until 2028), NATO (valid until 2029)
AI Security: Prompt injection prevention, adversarial robustness, model security
Data Privacy: PII detection/redaction, data anonymization, GDPR compliance
Governance: Model governance frameworks, AI ethics, bias detection

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NOTABLE AI PROJECTS & IMPLEMENTATIONS

Production RAG Systems

- Built enterprise-grade RAG systems processing 10M+ documents
- Implemented hybrid search combining semantic and keyword search
- Optimized retrieval pipelines reducing latency by 60%
- Custom chunking strategies improving retrieval accuracy by 40%

AI Agent Development

- Designed multi-agent systems for complex workflow automation
- Implemented ReAct and function-calling patterns for tool use
- Built autonomous agents with memory and planning capabilities
- Created agent orchestration frameworks for parallel execution

LLM Fine-tuning & Optimization

- Fine-tuned domain-specific models using LoRA/QLoRA
- Optimized prompts achieving 35% cost reduction while improving accuracy
- Implemented caching strategies reducing API costs by 50%
- Model quantization for edge deployment scenarios

AI Security Implementations

- Built prompt injection detection systems with 95% accuracy
- Implemented PII detection/redaction pipelines for sensitive data
- Created guardrails preventing model misuse and hallucinations
- Developed secure model serving infrastructure with encryption at rest/transit

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PROFESSIONAL EXPERIENCE HIGHLIGHTS

Founder & Lead AI Mentor / Delivery Pilot | 2026 – Present
Delivery Pilot (Skool Platform) · Remote

- Leading AI engineering mentorship and training on the Skool Platform to help professionals bridge skills gaps and become production-ready Forward Deployed Engineers (FDE)
- Designing curriculum across LLM orchestration, RAG architectures, prompt engineering, agentic workflows, and cloud-native AI delivery
- Conducting hands-on architecture reviews and portfolio project evaluations

Senior AI Engineer / DevOps Lead | 2020 - Present
Goldman Sachs, Ypsomed, Cushman & Wakefield

- Led AI transformation initiatives across finance, healthcare, and real estate sectors
- Architected production LLM systems serving millions of requests monthly

- Implemented MLOps pipelines reducing model deployment time from weeks to hours
- Mentored engineering teams on AI best practices and secure deployment patterns
- Established AI governance frameworks ensuring compliance and ethical AI use

Enterprise Transformation Architect | 2016 - 2020
Microsoft, Emerson, Various Fortune 500

- Led digital transformation initiatives incorporating AI/ML capabilities
- Built ML-powered automation systems increasing operational efficiency by 50%
- Designed cloud-native architectures supporting ML workloads at scale
- Evangelized AI adoption through workshops, training, and proof-of-concepts

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EDUCATION

🎓 **Bachelor of Science**
Southern New Hampshire University, USA 🇺🇸 | 2013

CERTIFICATIONS & CONTINUOUS LEARNING

📜 **Microsoft Certified Architect in Cloud Solutions** (70-532)
📜 **Azure AI Engineer Associate**
📜 **AWS Machine Learning Specialty**
📜 **Certified Kubernetes Administrator (CKA)**

Continuous Learning:
- Active contributor to AI/ML open-source projects - Regular participant in AI research paper discussions - Following latest developments in LLM research (GPT, Claude, Llama, Mistral) - Experimenting with cutting-edge AI techniques and frameworks

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SPEAKING & THOUGHT LEADERSHIP

- Technical blog on AI engineering best practices at rifaterdemsahin.com
- Internal tech talks on RAG architectures and LLM security
- Mentorship in AI/ML communities and engineering teams
- Code examples and tutorials shared via GitHub

SECURITY CLEARANCES

🔒 **UK SC (Security Check)** - Valid until 2028
🔒 **NATO Clearance** - Valid until 2029

✓ **Background Checks:** Watchdog (2024), Sterling (2019)

AVAILABILITY & CONTACT

Immediate Availability for contract or permanent AI engineering roles

📅 **Schedule a Discussion:** https://calendly.com/rifaterdem/schedule

📧 **Email:** contact@rifaterdemsahin.com

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SUPPORTING DOCUMENTS

📊 **Technical Portfolio & Presentations:**
https://rifaterdemsahin.com/wp-content/uploads/2025/02/rifaterdemsahinprofilepresentation.v2025.2.pdf

References and detailed project portfolios available upon request